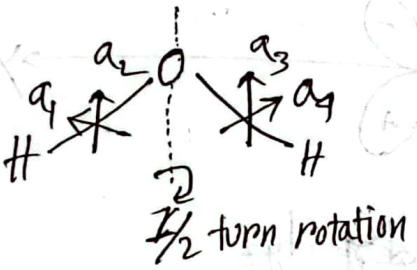
So, d_{xz} belongs to B_g representation.

TV, 2021

Generate a matrix of two fold rotation for H_2O using four arrows (a_2 & a_3 are perpendicular to the plane, a_1 & a_4 are init). use the same four arrows as the basis to matrix in the plane. Also find the characters of these matrix reppresentation?



⇒ By 1/2 turn rotation a_1, a_2, a_3, a_4 becomes,

$a_4, -a_3, -a_2, a_4$ respectively

the matrix representation

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} a_4 \\ -a_3 \\ -a_2 \\ a_1 \end{pmatrix}$$

~~The mir~~

∴ character = 0

By reflection in the plane: the matrix representation

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} a_1 \\ -a_2 \\ -a_3 \\ a_4 \end{pmatrix}$$

$\therefore$ character = 0.

TU, 2020, TU

⊛ Set up the molecular orbital diagram for CH_4, considering 2S & 2P orbitals of carbon and the 1S orbital of each hydrogen atom. Find hybridization of carbon atom? (T_d point group + point group এর character table দাও)

$\Rightarrow$ The structure of CH_4 molecule is tetrahedral.



By taking the four C-H bond as the basis, we can calculate the group orbitals for the four Hydrogen atom,

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the complete set of character of the reducible representation is,

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$
Γ	4	1	0	0	2

According to the character table of T_d point group,

No. of $A_1 = \frac{1}{24}[(4\times1\times1)+(1\times1\times8)+(0\times1\times3)+(0\times1\times6)+(2\times1\times6)]$

$= \frac{1}{24}(4+8+12) = 1$

No. of $A_2 = \frac{1}{24}[(4\times1\times1)+(1\times1\times8)+0+0+(2\times(-1)\times8)]$

$= 0$

No of $E = 0$

No of $T_1 = 0$

No of $T_2 = 1$

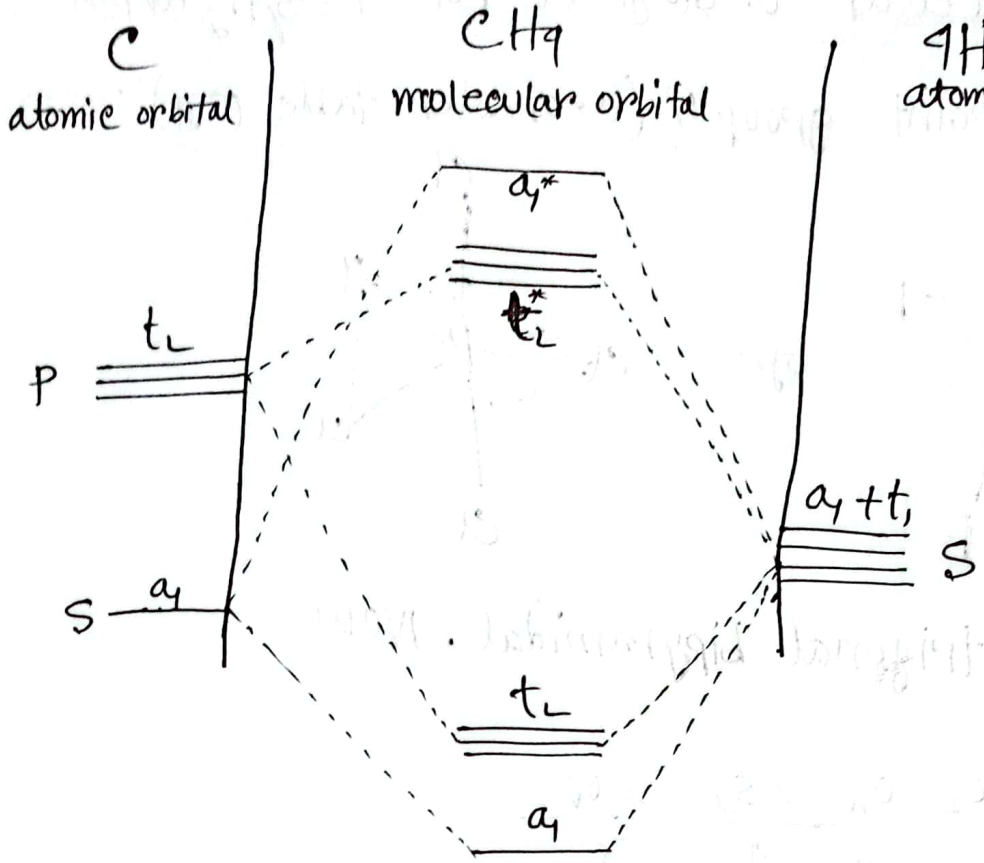
So we get, $\Gamma = A_1 + T_2$ (For 4 H atom)

From character table we can see that,

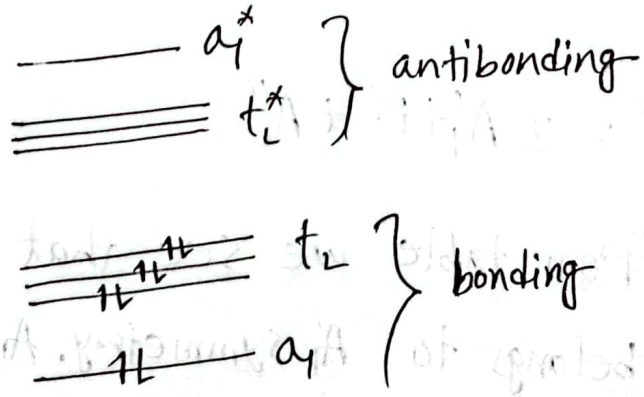
A_1 belongs to s orbital

T_2 belongs to P_x, P_y, P_z orbitals.

Molecular orbital diagram for CH_4:



Now, In CH_4 C atom has 4 valence electron & 4 H atom have 4 e^-. Now we can put these 4+4 = 8 e^- into the molecular orbital as follows.



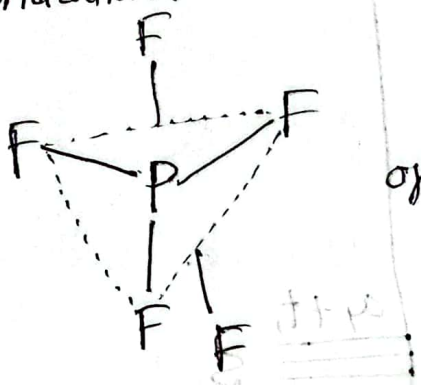
Bond order = $\frac{8-0}{2} = 4$

So there are 4 C-H bond. ~~No more bonding e⁻~~ & 4 pairs of e^- are occupy bonding orbitals.

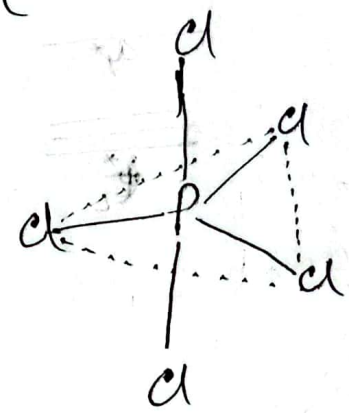
2021, Tu, 2022

Set up a molecular o. diagram for PCl_5/PF_5 which belongs to D_{3h} point group? (character table দেয়া) also find hybridization?

⇒



or



PCl_5 structure is trigonal bipyramidal. Now,

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ	5	2	1	3	0	3

Now,

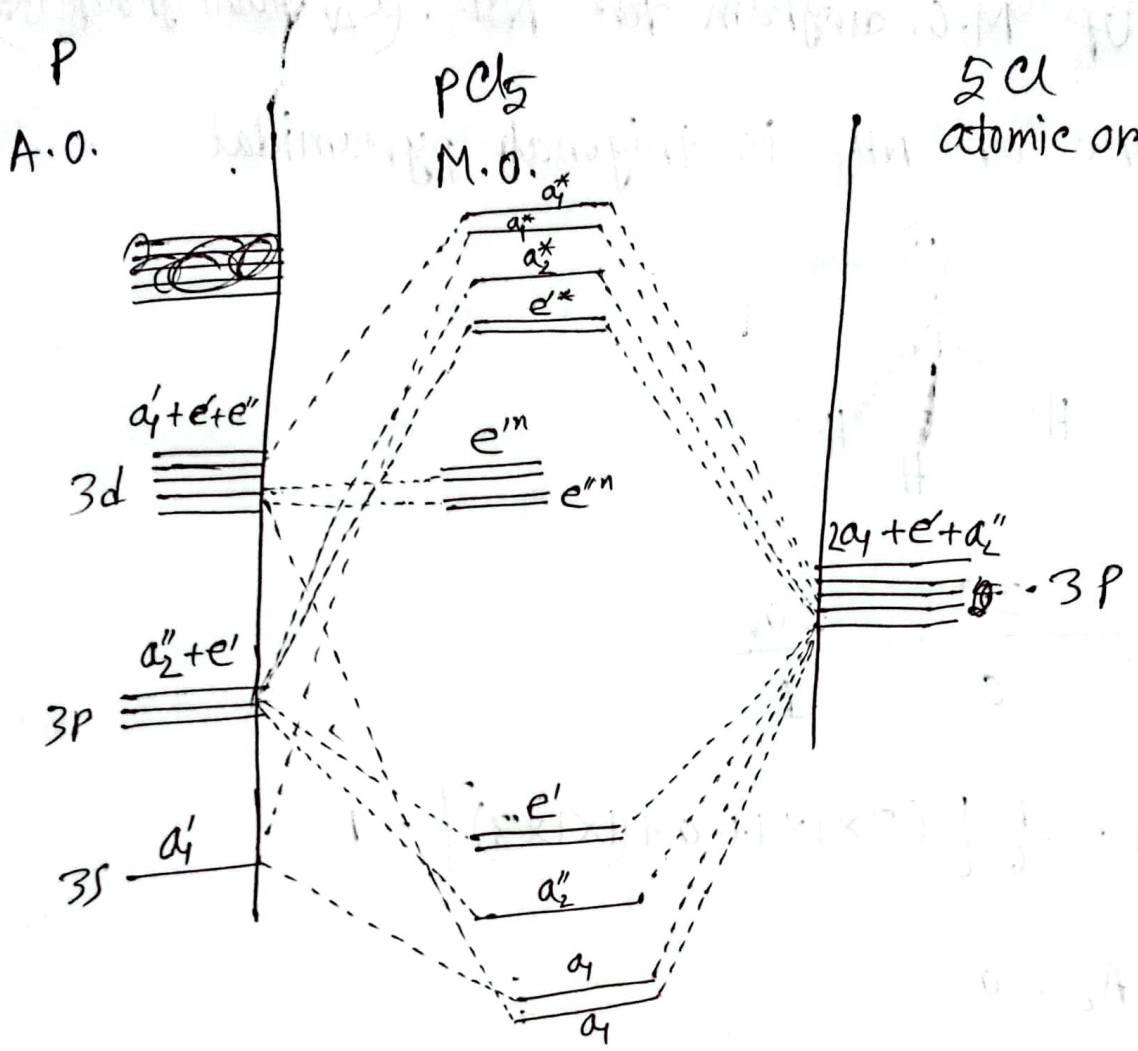
No. of $A_1' = \frac{1}{2}[(5\times1\times1)+(2\times1\times2)+(1\times1\times3)+(3\times1\times1)+0+(3\times1\times3)]$

$= 2$

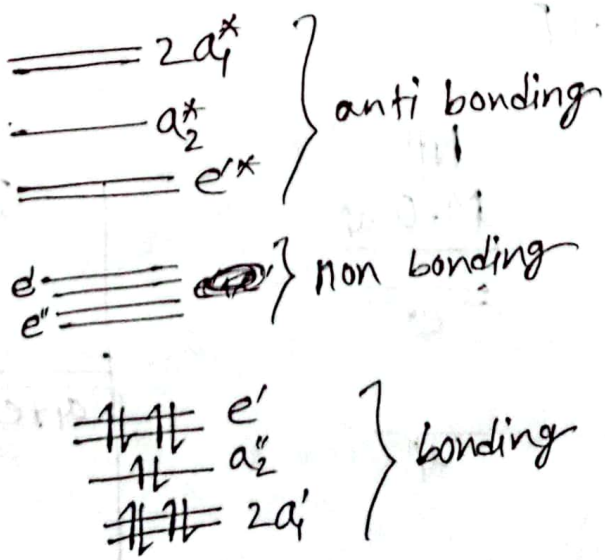
$A_2' = 0$, $E' = 1$, $A_1'' = 0$, $A_2'' = 1$, $E'' = 0$

so, we get, $\Gamma = 2A_1' + E' + A_2''$

Now, diagram: From table we see, that 's' orbital and d_{z^2} orbital belongs to A_1' symmetry. And p_x, p_y and $d_{x^2-y^2}$, d_{xy} belong in E' & p_z orbital belongs to A_2'' symmetry. so the hybridization is sp^3d



In PCl_5 there has total 10 e^- in the molecular orbital as follows:



Bond order = $\frac{10-0}{2} = 5$

(⊗ 2021, 2022 Set up M.O. diagram for NH_3? (C_{2v} point group) (5)

⇒ Structure of NH_3 is trigonal pyramidal



NOW,

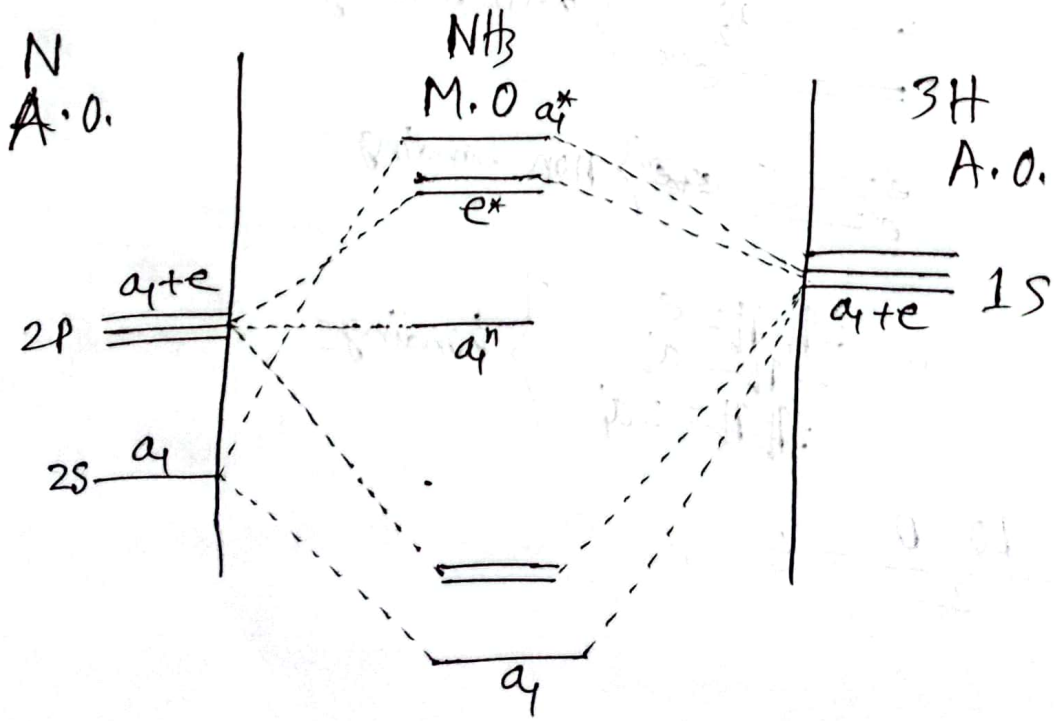
C_{2v}	E	$2C_3$	$3\sigma_v$
Γ	3	0	1

No. of $A_1 = \frac{1}{6}[(3\times1\times1+0+(1\times1\times3)] = 1$

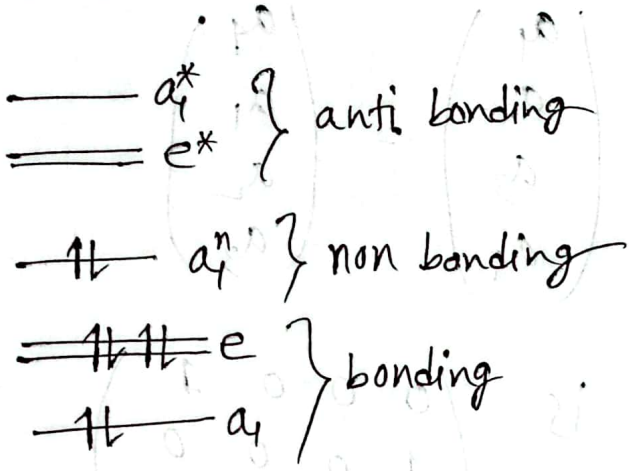
No. of $A_2 = 0$

No. of $E = 1$

$\therefore \Gamma = A_1 + E$

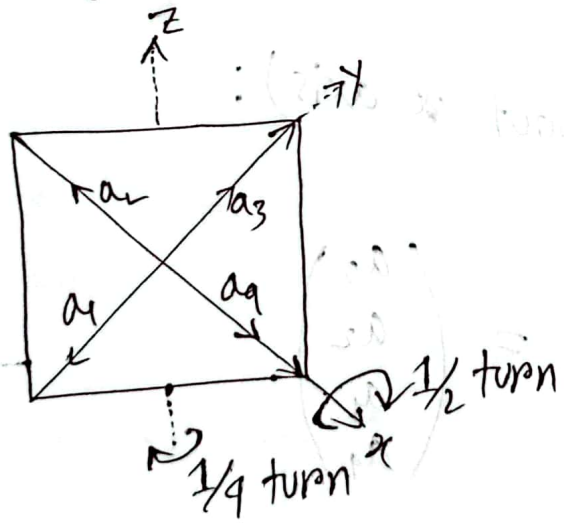


In NH_3, N has 5 valence e^- & 3H atom means total 8 e^- in valence shell in molecular orbital.



Bond order $= \frac{6-0}{2} = 3$.

✱ Use this diagram & Ans the question:



① Generate a matrix representation for quarter turn (about z axis) & half (about x axis) rotation?

② matrix for reflection in xy plane?

① For $\frac{1}{4}$ turn (about z axis)

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} a_4 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix}$$

So the matrix is $\begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$

and its character is = 0.

For 1/2 turn: (about x axis):

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} a_3 \\ a_2 \\ a_1 \\ a_4 \end{pmatrix}$$

So the matrix is $\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ & character = 2

(ii) For reflection in xy plane ?:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}.$$

So the matrix is $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ & character = 4

2020

(iii) Set up the matrices which will perform the following transformations:

(A) $\begin{pmatrix} x \\ y \end{pmatrix}$ to $\begin{pmatrix} -y \\ -x \end{pmatrix}$ (B) $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ to $\begin{pmatrix} -y \\ x \\ -z \end{pmatrix}$?

(A) The full set of eqn is,

$$x = (0 \times x) + (-1 \times y)$$

$$y = (-1 \times x) + (0 \times y)$$

The matrics can be written down as $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$.

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(B) The full set of eqn is,

$$x = (0 \times x) + (-1 \times y) + (0 \times z)$$

$$y = (1 \times x) + (0 \times y) + (0 \times z)$$

$$z = (0 \times x) + (0 \times y) + (-1 \times z)$$

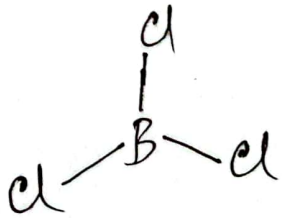
The matrix can be written down by the inspection as,

$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

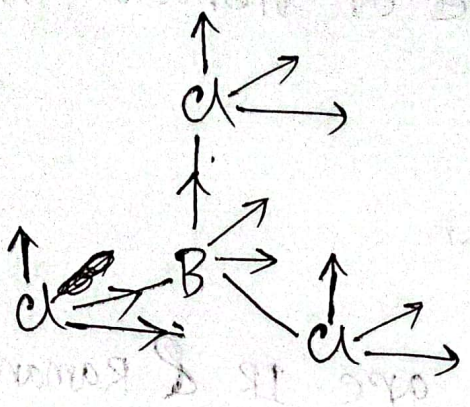
2021

Find the number & symmetry species of the Raman & IR active vibration of boron trichloride (BCl_3) (character table D_{3h}) ?

⇒ Structure of BCl_3 is trigonal planar.



Here N=4, the basis of the representation is the set of the $3 \times 4 = 12$ arrows.



Now,

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ	12	0	-2	4	-2	0

No of $A_1' = \frac{1}{12}[(12\times1\times1)+0+(-2\times1\times3)+(4\times1\times1)+(-2\times1\times2)+0]$

$= 1$

Similarly, $A_2' = 1$, $E' = 3$, $A'' = 0$, $A_2'' = 2$, $E'' = 1$

So, $\Gamma = A_1' + A_2' + 3E' + 2A_2'' + E''$

There are total ~~movement~~ molecular movement $= 12$.

Total movement: $A_1' + A_2' + 3E' + 2A_2'' + E''$

Rotations : A_2' $+ E''$

Translation : $E' + A_2''$

$\therefore$ Vibration $= A_1' + 2E' + A_2''$

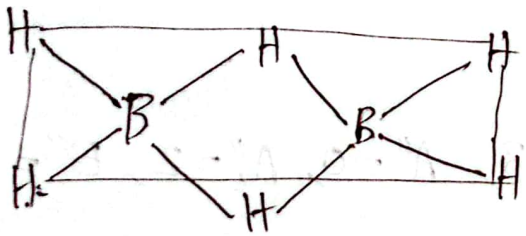
There are total 6 mode of vibration is possible.

IR active = $2E' + A_2''$

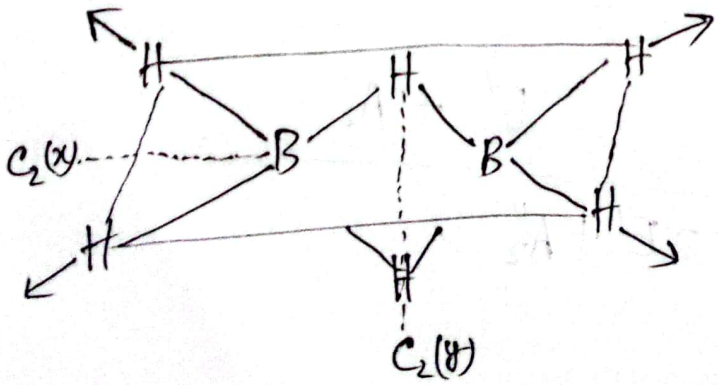
Raman active = $A_1' + 2E'$

There are 5 vibration are IR & Raman active.

⊛ 2022 Find the number of terminal B-H stretching vibrations which are active in the IR & Raman spectra of diborane (B_2H_6). (D_{3h} point group (নয়))



⇒ Here total number of terminal B-H bond is 4. The basis of the representation is the set of the 4 arrows.



Now, the complete set of the character of the representation is,

D_{2h}	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$
Γ	9	0	0	0	0	9	0	0

No of $A_g = \frac{1}{8}[9\times1\times1+0+0+0+0+9\times1\times1+0+0] = 1$

Similarly, $B_{1g} = 1$, $B_{2g} = 0$, $B_{3g} = 0$, $A_u = 0$, $B_{1u} = 0$, $B_{2u} = 1$, $B_{3u} = 1$

our basis is only included stetching of B-H. we do not need to remove translation or rotation because we did put them in when setting up the basis of the representation.

IR active $\longrightarrow B_{2u} + B_{3u}$

Raman active $\longrightarrow A_g + B_{1g}$

So the number of terminal B-H stretching vibrations which are Raman & IR active both ar 2.

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